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# 1.1 What is Claude Code

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## 1.1.1 Definition of Claude Code

Claude Code is an intelligent coding assistant launched by Anthropic for developers. It is not just a simple code generation tool, but an "Agentic AI" system with memory, tool calling, autonomous planning, and environmental awareness capabilities. Through deep integration with local development environments (such as VS Code), Claude Code can deeply understand codebases, explain complex architectures, generate visualization diagrams, and execute actual code modification tasks without leaving the safety boundary.

### 📌 Distinguishing Claude Code from the Claude LLM

Claude Code is a tool software and cannot be equated with the Claude LLM. Although the official team of Claude Code uses the Claude LLM by default within the tool, we can configure it to use non-official large models, such as domestic GLM models.

This also demonstrates the flexibility and scalability of AI assistants, allowing developers to choose the most suitable model according to their needs and scenarios.

Although currently among the programming models on the market, Claude 4.5 Opus is the smartest and most capable, it is too expensive, so in actual development, we mainly use domestic models like GLM.

## 1.1.2 Development Background of Claude Code

In the field of AI-assisted programming, technology development has gone through three main stages, from early code completion tools (such as TabNine, GitHub Copilot) to assistants with simple conversational capabilities, and now to intelligent agent systems:

1. **Code completion stage:** Provides simple code snippet completion based on statistical models
2. **Conversational assistance stage:** Combined with large language models, capable of understanding natural language instructions and generating code

3. **Intelligent agent stage:** Has autonomous planning, tool calling, environmental awareness, and memory capabilities, capable of completing complex development tasks

Claude Code is a representative product in the third stage, and its emergence marks the transition of AI programming assistants from "passive response" to "proactive collaboration."

### 1.1.3 Core Positioning of Claude Code

The core positioning of Claude Code is "a smart collaboration partner for developers." It aims to:

- **Enhance developer capabilities:** Improve development efficiency and code quality through AI assistance
- **Simplify complex tasks:** Break down complex development tasks into executable subtasks
- **Provide deep insights:** Offer architecture-level suggestions based on comprehensive understanding of the codebase
- **Ensure safety and reliability:** Operate within safety boundaries to protect code and data privacy
- **Support team collaboration:** Provide shared knowledge and context to facilitate team collaboration

### 1.1.4 Technical Foundation of Claude Code

Claude Code is built on Anthropic's Claude large language model, combined with the following core technologies:

1. **Large language model:** Based on the Claude Opus 4 and Sonnet 4 dual-model system, with powerful natural language understanding and generation capabilities
2. **Tool calling system:** Capable of calling various built-in and external tools to extend AI capabilities
3. **Memory system:** Has short-term and long-term memory capabilities, capable of remembering context and historical interactions
4. **Autonomous planning algorithm:** Capable of breaking down complex tasks into executable subtask sequences
5. **Environmental awareness capability:** Capable of perceiving and understanding the local development environment, including codebase structure and file content
6. **Security mechanism:** Based on the principle of least privilege, all operations are executed within safety boundaries

### 1.1.5 Main Components of Claude Code

Claude Code mainly consists of the following core components:

| Component                 | Function                                                                                 |
|---------------------------|------------------------------------------------------------------------------------------|
| Agentic Planner           | Core controller, responsible for converting user requests into executable task sequences |
| Memory System             | Stores and manages context information, historical interactions, and project knowledge   |
| Tool System               | Provides built-in tool and external tool calling capabilities                            |
| Code Understanding Engine | Deeply understands codebase structure and content                                        |
| Execution Engine          | Executes actual code modifications and tool calls                                        |
| Security Manager          | Ensures all operations are executed within safety boundaries                             |

### 1.1.6 Essential Differences Between Claude Code and Traditional AI Programming Tools

Compared with traditional AI programming tools, Claude Code has the following essential differences:

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1. Project-level understanding: Capable of understanding the structure and relationships of the entire codebase, not just individual files
2. Autonomous planning capability: Capable of autonomously decomposing tasks, formulating plans, and executing them
3. Tool calling capability: Capable of calling various tools to extend its capabilities
4. Environmental awareness capability: Capable of perceiving and adapting to the local development environment
5. Memory capability: Capable of remembering context and historical interactions
6. Safety boundary: Operates within safety boundaries to protect code and data privacy

### 1.1.7 Design Principles of Claude Code

Claude Code follows these core principles:

1. Safety first: All operations are executed within safety boundaries to protect user code and data

2. **Least privilege:** Tool calling and code modification follow the principle of least privilege, requiring user confirmation
3. **Transparent and trustworthy:** All operations are transparent to users, who can intervene and control at any time
4. **Enhanced collaboration:** Aims to enhance developer capabilities, not replace developers
5. **Continuous learning:** Capable of learning from interactions and continuously improving its capabilities
6. **Open ecosystem:** Supports plugin extensions and encourages community contributions

### 1.1.8 Application Prospects of Claude Code

With the continuous development of AI technology, Claude Code, as a representative of intelligent agent systems, has broad application prospects:

1. **Improve development efficiency:** Automate repetitive tasks, allowing developers to focus on creative work
2. **Lower development barriers:** Help beginners get started quickly and reduce the learning curve for programming
3. **Promote technology dissemination:** Promote the dissemination of technical knowledge through code explanation and documentation generation
4. **Accelerate innovation:** Help developers quickly validate ideas and accelerate product innovation
5. **Change development models:** Promote the shift from individual development to human-machine collaborative development models

Claude Code is not just a tool, but the future direction of AI-assisted development. It will redefine the collaborative relationship between developers and AI, ushering in a new era of intelligent development.

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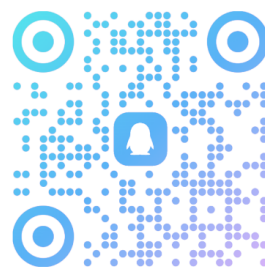
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